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3D-PRINTED SOFT MAGNETIC CORES FOR COMPACT ELECTROMECHANICAL DEVICES VIA MATERIAL EXTRUSION

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ABSTRACT

This paper reports the fabrication and characterization of 3D-printed, soft magnetic cores manufactured via material extrusion that attain a relative magnetic permeability above 30. The cores are monolithically fabricated from pellets of nylon-12 doped with FeSiAl particles and do not require heat postprocessing, which makes them compatible with polymer-based, multi-material fabrication. The proposed cores exhibit a very low magnetic retentivity and negligible electrical conductivity, and they outperform the cores of similar characteristics found in the literature by an order of magnitude. This work is of great interest for the inexpensive, single-step fabrication of monolithic electromechanical devices.

KEYWORDS

3D-printed magnetics, material extrusion, additive manufacturing, 3D-printed electromechanical devices.

INTRODUCTION

Electrical machines (e.g., generators, motors, transformers) are the key elements in the generation, distribution, and utilization of electrical energy. The ever-growing energy needs of modern society and the necessary transition to clean, electricity-based forms of energy utilization (e.g., electric vehicles) motivate the search for strategies to enhance the performance of electrical machines. Recent reported work demonstrates that additive manufacturing (AM) technologies have the capability to produce all the parts involved in the fabrication of electrical machines [1] including magnetic cores [2]-[7], conductive coils [8],[9], permanent magnets [10],[11] and mechanical couplings [12]. By harnessing the virtues of AM (e.g., multi-material fabrication, waste reduction, feasibility to create freeform, geometrically complex structures), the fabrication of electrical machines can become cheaper and easier, and their performance can be improved [13].

The performance of electrical machines is closely tied to that of their magnetic cores, which are made of soft magnetic materials. Multiple studies have reported the development of AM-compatible soft magnetic feedstock [14]. However, most of the proposed materials require heat postprocessing [2]-[5], which is incompatible with thermoplastics-based, multi-material 3D printing. This circumstance significantly dampens the potential benefits of using AM. Moreover, sintering, often carried out at 950 ºC or higher temperatures, is a lengthy process that is prone to geometrical distortion of the fabricated components. While 3D-printed, soft magnetic parts that do not require postprocessing exist, their reported performance is modest (i.e., relative magnetic permeabilities below 10) [6],[7],[15],[16]. This work reports soft magnetic cores that do not require heat postprocessing and that attain relative magnetic permeabilities above 30. The cores are fabricated in a single step using a multi-material extrusion 3D printer capable of handling other functional materials (e.g., electrically conductive, elastic). This feat entails significant progress towards the monolithic additive manufacture of electrical machines, and it is of immediate relevance for the on-site fabrication of electromechanical devices in remote areas with limited access to standard manufacturing techniques.

METHODS

Manufacturing Technology: Material Extrusion

The cores reported in this work are fabricated via material extrusion using pellets as feedstock. Material extrusion is an established AM technology that creates parts by pushing feedstock through a nozzle and depositing it layer by layer, building parts from the ground up. Material extrusion is one of the most accessible and affordable 3D printing technologies, and it is one of the few AM techniques that seamlessly allows monolithic multi-material fabrication, which is essential for the single-step 3D-printing of electrical machines [13]. The 3D printer used in this study is an E3D Motion System and ToolChanger (E3D-Online, Chalgrove, Oxfordshire, United Kingdom) customized to accommodate a Mahor v4 pellet extruder (Mahor.xyz, Spain) [8].

Materials: FeSiAl-doped Nylon-12 Pellets

Feedstock for material extrusion is most often found in filament form. However, the amount of functional filler that a filament can contain before it becomes too brittle to print with is only about 55 % in weight [10]. Because of this limit, soft magnetic parts printed out of filament perform poorly. Other forms of feedstock, such as paste [2] and pellets [8], allow higher filler content. While parts printed from paste usually require sintering to become mechanically stable, parts printed from pellets can be used as printed. Moreover, the use of feedstock in the form of pellets allows access to a wide range of materials, including some originally developed for industry-grade, polymer-based processes (e.g., injection molding). Therefore, parts printed from pellets have the potential to yield better performance than their filament-based counterparts, while not requiring post-processing. The magnetic cores presented in this work are printed from pellets of nylon 12 doped with FeSiAl particles at 86 % in weight (MATE Co., Okayama, Japan).

Sample Preparation: Toroidal cores

To evaluate the potential of the proposed process, two sets of toroidal cores were 3D printed (Fig. 1). One set was designed to fit inside a Keysight 16454A magnetic material test fixture (Keysight Technologies, Santa Rosa, CA USA), while the other set was designed for comfortable winding

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(Fig. 2). The dimensions of each set of cores are shown in Table 1, and the key printing settings used are summarized in Table 2. The "extrusion multiplier" (EM) parameter regulates the rate at which the material is pushed through the nozzle, controlling the density and consistency of the printed parts. The EM was varied between 0.5 and 1.5 to evaluate its effect on the magnetic performance of the 3D-printed cores.

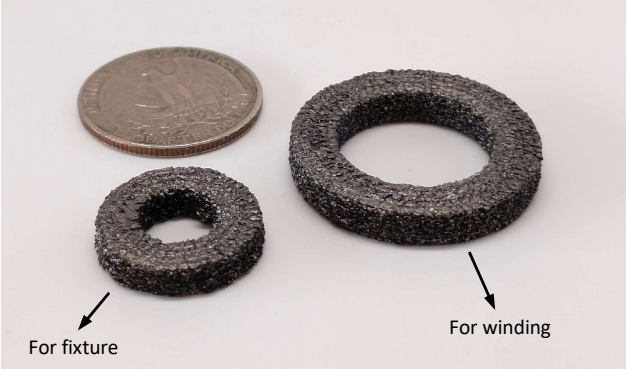


Figure 1. 3D-printed toroidal cores of different sizes made in FeSiAl-doped nylon next to a US quarter.



Figure 2. Toroidal core with two enameled wires (22 and 28 AWG) wound around it, forming a transformer.

Table 1: Nominal dimensions of the 3D-printed cores.

Core set	For fixture	For winding
Inner diameter (mm)	10	20
Outer diameter (mm)	18	30
Height (mm)	4	5

Table 2: Key 3D printing parameters.

Nozzle diameter (mm)	0.8
Extrusion multiplier (-)	0.5 – 1.5
Layer height (mm)	0.2
Extrusion temperature (ºC)	270
Bed temperature (ºC)	90
Speed (mm/min)	1000

Relative Magnetic Permeability Measurement

The relative magnetic permeability of the samples was estimated using two approaches:

A. The wound cores (Fig. 2) were connected to an 895 LCR Meter (B&K Precision, Yorba Linda, CA USA) to extract values of series inductance and resistance between 100 Hz and 100 kHz. The series inductance measurements were used to approximate the relative permeability of each sample according to Eq 1.

$$\mu_r' = \frac{2\pi L_s}{\mu_0 N^2 h \ln\frac{c}{b}} \quad (1)$$

where μ_r' is the real component of the relative permeability of the sample (the imaginary component is assumed negligible in this range of frequencies), L_s is the measured series inductance, N is the number of turns on the winding, and h, b, and c are the measured height, inner diameter, and outer diameter of the sample, respectively.

B. The smaller cores (Fig. 1, left) were placed inside the 16454A magnetic material test fixture. Using the HP42942A terminal adapter, this fixture was connected to a Keysight 4294A Precision Impedance Analyzer. This setup was used to obtain measurements of impedance from 100 kHz to 10 MHz, from which the complex relative permeability was computed as follows [17]:

$$\mu_r^* = \mu_r' - j\mu_r'' = \frac{2\pi(Z_m^* - Z_{sm}^*)}{j\omega\mu_0 h \ln\frac{c}{b}} + 1 \quad (2)$$

where μ_r^* is the complex relative permeability of the sample, Z_m^* is the complex impedance measured for the sample, Z_{sm}^* is the complex impedance measured while the fixture is empty, and h, b, and c are the measured height, inner diameter, and outer diameter of the sample, respectively.

B-H Characterization Circuit

A custom electronic circuit was built to obtain the magnetization (B-H) curves of the wound cores (Fig. 3). The circuit drives the primary winding of a toroidal transformer via a voltage-controlled current source (VCCS) to generate a magnetic field (H), and integrates the voltage induced on the secondary winding to enable measurement of the magnetic flux density inside the core (B). A TIP142 and TIP147 Darlington transistor complementary pair, controlled by an OPA552 operational amplifier, form the VCCS. A 0.5-ohm sense resistor and a differential amplifier using two 100 kΩ resistors measure the current supplied to the primary side. An active integrator consisting of a 1 kΩ input resistor, a 330 nF feedback capacitor, and a 1 MΩ feedback resistor to prevent DC integration, integrates the secondary-side voltage. Both the differential amplifier and the integrator utilize ADTL082 operational amplifiers. A Keysight DSOX4145A oscilloscope provides the input signal for the VCCS and measures the output voltages of the circuit.

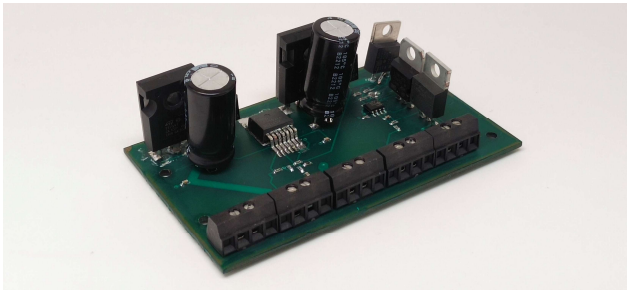


Figure 3. Picture of the electronic circuit used for B-H characterization of the wound cores.

RESULTS

The 3D printed cores reach relative magnetic permeabilities above 30 for values of EM equal to or larger than 1. As EM decreases, so does the permeability of the 3D-printed cores. Figure 4 shows the measurements of relative complex magnetic permeability versus frequency for each 3D-printed core sample. The solid lines depict the real component of the relative permeability, while the

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dotted lines show its imaginary component. The LCR-meter setup was used to obtain data at frequencies between 100 Hz and 100 kHz, for cores with EM = 0.75, 1.0 and 1.25, one sample each. Similarly, the impedance analyzer setup was used to collect data between 100 kHz and 10 MHz, for cores with EM = 0.5, 0.75, 1.0, 1.25, 1.5, three samples each. The difference in the number and diversity of the samples is due to the time-consuming process of hand winding cores. A discontinuity of 5-10 units is appreciated between the data coming from either measuring setup. The lowest values, given by the impedance analyzer setup, are assumed more accurate.

Figure 5 depicts the real component of the relative magnetic permeability at 1 MHz, as measured by the impedance analyzer, versus the EM parameter.

Figure 6 shows the B-H curve of a wound, 3D-printed core with EM = 1.0; the core weights 5.09 grams and approximately occupies 2.43 cm^3. The data were obtained by exciting the primary winding with an alternating current of 2.5 amperes root mean square (RMS) at 100 Hz, via the previously described setup. The B-H curve reveals that the 3D-printed cores exhibit a narrow hysteresis loop with a modest coercive field (below 200 A/m) and very low retentivity (less than 10 mT). The cores saturate slowly for flux densities above 0.15 T.

DISCUSSION

The data shown in Figs. 4 and 5 show that the cores fabricated in this study can attain magnetic permeabilities above 30. This value of permeability is significantly superior to those reported by other works on 3D-printed soft magnetic cores that do not involve heat post-processing, and it is on par with the permeability reached by 3D-printed cores that rely on moderate-temperature sintering (Table 3). Other reported 3D-printed cores post-processed at high temperatures (above 900 ºC) reach permeabilities about one order of magnitude higher, at the cost of losing compatibility with polymer-based 3D-printing processes.

Figure 5 demonstrates that by tuning 3D printing parameters, the magnetic performance of the fabricated devices can be modulated. In particular, the EM parameter, which effectively translates to density, allows production of parts with a similar geometry, but different magnetic permeability. Other reports have characterized similar phenomena involving additional 3D-printing parameters [18]. Figure 5 also shows that for EM values below 1 (i.e., sub-extrusion), the relative permeability of the cores depends linearly on EM, while, for values of EM above 1 (i.e., over-extrusion), the relative permeability saturates. This is motivated by the fact that the fabricated parts reach their full density at EM = 1. Further increasing EM does not translate to higher density, but it instead results in material overflow and increased part volume. The ability to obtain a range of relative permeabilities enables the monolithic fabrication of multi-permeability inductors, which can yield higher inductance values and efficiencies than their single-permeability counterparts, by increasing the uniformity of the magnetic flux within the core [19],[20]. This is especially useful in high-frequency applications, where lower permeability materials are preferred due to their smaller losses.

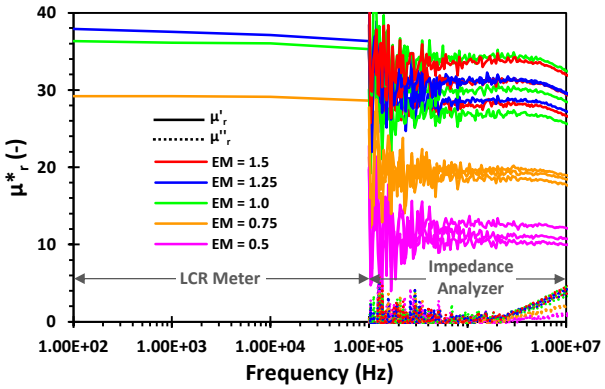


Figure 4. Relative complex magnetic permeability of 3D-printed cores versus frequency, for different EM values.

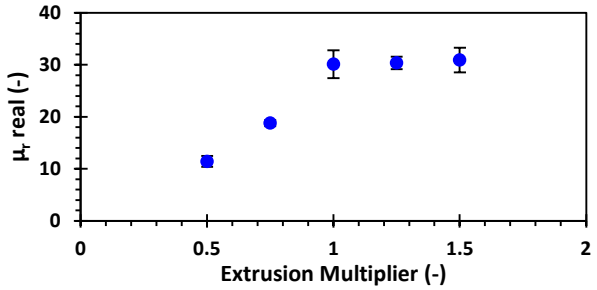


Figure 5. Real component of the relative magnetic permeability of 3D-printed cores at 1 MHz versus EM.

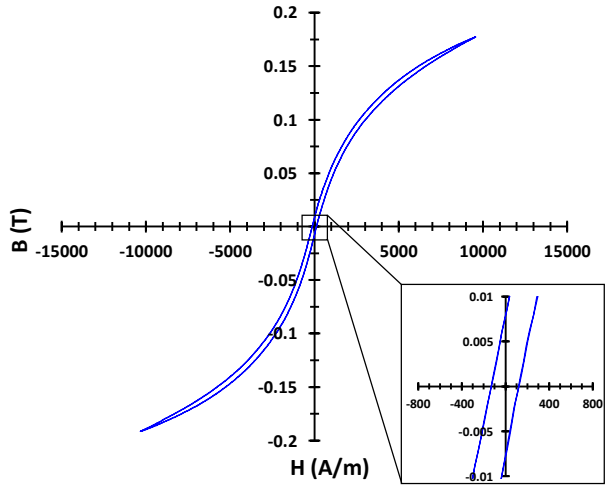


Figure 6. B-H curve obtained from a wound 3D-printed core with EM = 1.0.

Table 3: Comparison of relative permeabilities of 3D-printed soft magnetic components.

Work	μ_r	Requires sintering [temperature]
This work	31	No
Bollig, 2017 [6]	~1.4*	No
Bowen, 2019 [7]	3.9-8.9**	No
Díaz-García, 2022 [15]	~1.9*	No
Khatri, 2018 [16]	~1.6*	No
Ding, 2018 [2]	35	Yes [200 ºC]
Liu, 2018 [3]	83	Yes [950 ºC]
Liu, 2018 [3]	103	Yes [1000 ºC]
Trnka, 2020 [4]	~600*	Yes [unspecified]
Liu, 2020 [5]	495	Yes [950 ºC]

* Not stated, estimated from related data.

** The authors state that 3.9 is the most reliable value.

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The B-H data (Fig. 6) show that the 3D-printed cores can sustain a considerable magnetic flux density before saturating, confirming their suitability for the fabrication of electrical machines.

The proposed process enables the use of high-performance soft magnetic materials while remaining compatible with polymer-based multi-material 3D printing. In conjunction with the use of other functional materials (e.g., electrically conductive, hard magnetic, elastic), this technology has the potential to enable the single-step fabrication of compact electrical machines and electromechanical devices (e.g., robots).

CONCLUSION

This work reported the fabrication and characterization of 3D-printed soft magnetic cores that do not require heat prost-processing and that can attain relative permeabilities above 30. The reported cores exhibit a narrow hysteresis loop with a retentivity of less than 10 mT and negligible electrical conductivity. The proposed process enables the fabrication of multi-permeability components, and it is compatible with most multi-material 3D-printing technologies, making it suitable for the single-step fabrication of 3D-printed electromechanical devices.

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